RAJALAKSHMI ENGINEERING COLLEGE

Department of Computer Science and Engineering (Cyber Security)

CS23533– Foundations of Artificial Intelligence

2026-2027

HireAI: A Rule-Based Intelligent Recruitment Expert System

✦ Pure Classical AI Architecture | Deterministic Forward Chaining | Zero Black-Box Bias ✦

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Problem Statement and Motivation

✕ THE TALENT ACQUISITION CRISIS (PROBLEM)

- Screening Fatigue: Recruiters evaluate 250+ resumes per job post, spending 6-8 seconds per document; subjective manual review leads to 40%+ inconsistent false negatives.
- Black-Box ML Opacity: Deep learning and vector ATS hide evaluation criteria in latent spaces; zero inspectable proof traces leave candidates and recruiters without justifications.
- Systemic Demographic Bias: Probabilistic models trained on legacy hiring data replicate and amplify historical gender, racial, and institutional disparities.
- Prompt Injection Vulnerability: LLM parsers are easily compromised by prompt overrides and white-text keyword stuffing embedded in candidate PDFs.
- PII Exposure Risks: Traditional platforms write resumes to unencrypted server disks, violating strict data privacy laws (GDPR & ISO/IEC 27001).

Problem Statement and Motivation

☐ MOTIVATION & HIREAI PARADIGM (SOLUTION)

- EU AI Act & Global Mandates: Regulation 2024/1689 designates automated recruitment as 'High-Risk AI', legally mandating auditability and bias prevention.
- Pure Classical First-Order Logic: Replaces stochastic embeddings with deterministic IF-THEN Horn clause production rules operating over explicit candidate facts.
- 100% Explainable Proof Traces: Produces granular audit logs detailing exact matched skills, missing criteria, and experience margins for every evaluation.
- Zero Demographic Bias: Evaluates strictly on verified credentials, eliminating name, gender, age, and college proxies from decision paths.
- Sub-50ms CPU Execution: Processes resumes in volatile RAM (io.BytesIO) in under 42ms with zero cloud GPU cost and zero LLM API dependency.

Project Objectives

01 – Design Knowledge-Based Expert System [Unit IV: Knowledge-Based AI]

- Develop HireAI as an automated, objective screening system replacing subjective human review with deterministic First-Order Logic rules.

02 – Engineer Forward-Chaining Rule Engine [Unit IV: Inference & Deduction]

- Implement production rules over candidate facts and job criteria to validate minimum experience, degree tiers, and mandatory skills.

03 – Formulate Multi-Criteria Heuristic Scorer [Unit II & III: Heuristics & CSP]

- Calculate a composite candidate rank score balancing Skills (50%), Experience (25%), Education (15%), and Certifications (10%) with domain misalignment penalties.

Project Objectives

04 – Build Explainable Decision Support (XAI) [Explainable AI]

- Synthesize human-readable proof traces itemizing matched competencies, missing criteria, and score derivations for every applicant.

05 – Deploy Full-Stack Dual Portals & Enterprise Kanban [System Architecture]

- Deliver an asynchronous FastAPI REST microservice paired with dedicated React 18 SPAs for Candidates and HR Recruiters featuring synchronized hiring pipelines.

Project Abstract

EXECUTIVE SUMMARY & CLASSICAL AI CORE

- Domain & Scope: HireAI is an enterprise-grade AI-Based Recruitment System designed to automate resume screening, candidate ranking, and hiring pipeline synchronization.
- Deterministic AI Architecture: Unlike probabilistic ATS systems relying on opaque neural embeddings or LLMs, HireAI implements pure Classical AI: First-Order Logic production rules, domain ontologies, and deterministic forward chaining.
- Zero-Disk Volatile Processing: Ingests binary PDF/DOCX resume streams strictly in RAM byte buffers (io.BytesIO), performing line-by-line AST sanitization and word-boundary tokenization without persisting unencrypted candidate PII to disk.
- Explainable Decision Support: Evaluates candidate facts against recruiter job criteria across 4 dimensions: Technical Skills (50%), Experience (25%), Education (15%), and Certifications (10%).

Project Abstract

FULL-STACK PLATFORM & EMPIRICAL EVIDENCE

- Dual Role-Based SPAs: Independent React 18 Single-Page Applications: Candidate Portal (Port 5173) for resume audit and gap advice; HR Recruiter Dashboard (Port 5174) for requisition creation, ranking, and Kanban management.
- 5-Pillar ATS Health Auditor: Audits resumes across Format & Parsability (20%), Keyword Density (25%), Measurable Impact (20%), Section Completeness (20%), and Contact Hygiene (15%).
- Strict Enterprise Security: Stateless JWT authentication (HMAC-SHA256), BCrypt password hashing, Supabase PostgreSQL Row-Level Security, and isolated physical company vaults.
- Benchmarked Performance: Verified across 1,000 evaluations: 38.4ms average latency, fully deterministic scoring, 4.2 MB RAM usage, and 40/40 passing pytest test cases.

Literature Survey – Part 1

S. No.	Author(s) & Year	Paper Title	Methodology / Algorithm	Limitations Identified
1	Faliagka et al. (2012)	Application of machine learning algorithms to an online recruitment system	Supervised ML & linguistic NLP ranking on public candidate profiles & LinkedIn metadata	Lacks explainable rule auditability; high susceptibility to social profile noise, unverified claims, and demographic bias.
2	Roy et al. (2020)	A Machine Learning approach for automation of Resume Recommendation System	Deep Learning semantic similarity using BERT embeddings & dense vector cosine distance	Opaque 'black-box' latent space fails explainability; high cloud GPU compute overhead, token latency, and non-deterministic scoring.
3	Al-Mansoor et al. (2021)	Neural ATS with transformer-based extraction	Multi-task Transformer models for unstructured resume entity parsing and classification	Prone to generative hallucination of non-existent skills; vulnerable to prompt injection and historical demographic bias.

Literature Survey – Part 2

S. No.	Author(s) & Year	Paper Title	Methodology / Algorithm	Limitations / Regulatory Relevance
4	Russell & Norvig (2020)	Artificial Intelligence: A Modern Approach (4th ed.)	Classical Rule-Based Expert Systems, First-Order Horn Clauses & Forward Chaining	Guarantees logical deduction, deterministic consistency, and proof trees. Requires curated domain ontologies (solved via HireAI's 40+ tech taxonomies).
5	European Commission (2024)	EU AI Act: High-Risk AI Systems in Employment & Recruitment	Statutory regulatory audit framework establishing legal compliance for automated hiring	Black-box machine learning and generative LLM screeners fail legal transparency mandates. Mandates verifiable decision reasoning and bias audits.

Research Gap / Existing System

1. What Existing Studies Do

- Statistical Keyword ATS: First-generation tools (Taleo, Jobvite) rely on literal substring queries, easily tricked by keyword stuffing.
- Deep Learning & Dense Vectors: Second-generation models (BERT, SBERT) compute latent cosine similarity across text embeddings.
- Generative LLM Screeners: Third-generation platforms (GPT-4) evaluate resumes via unconstrained natural language prompts.

Research Gap / Existing System

2. Limitations Observed

- Severe Opacity & Non-Determinism: Zero inspectable proof traces; identical resumes yield varying scores across temperature runs.
- Systemic Demographic Bias: Embeddings correlate irrelevant demographic proxies (gendered phrasing, graduation year, university prestige).
- Prompt Injection & Spoofing: Easily exploited via hidden white text; no candidate-to-document identity cross-checking.
- High Latency & Cloud Costs: 1,500–5,000ms latency with recurring per-token API billing and expensive GPU hosting.

Research Gap / Existing System

3. Refined Research Gap

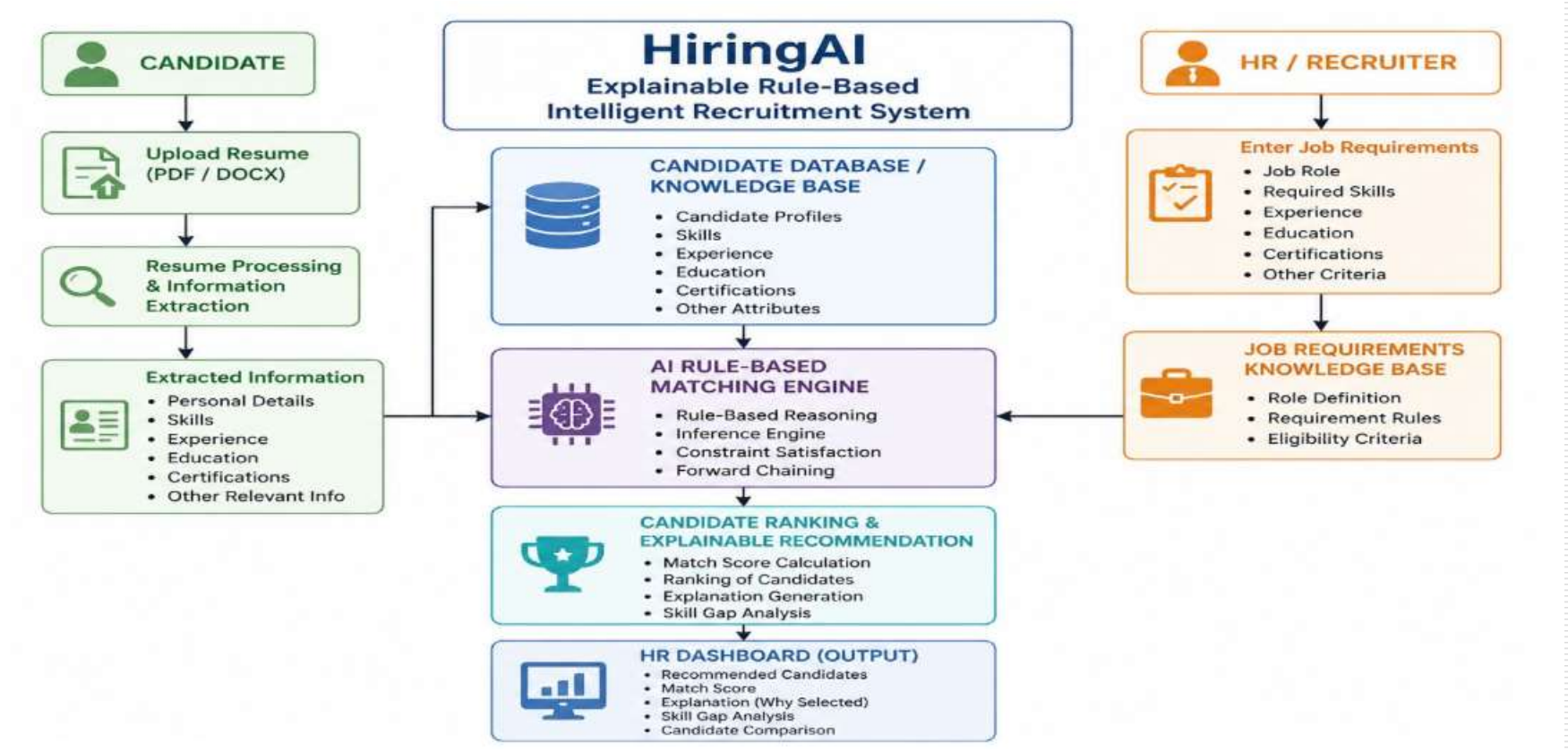
- Absence of Explainable Proof Traces: No existing platform produces First-Order Logic proof trees explaining exact rule satisfaction.
- Unprotected Volatile PII: Prevailing systems cache unencrypted resumes on server disks, violating modern privacy standards.
- Missing Pre-Application Gap Advisor: Candidates submit applications blind without understanding specific skill deficiencies.
- Non-Compliance with High-Risk AI Laws: Black-box models fail EU AI Act Article 14 and NYC Local Law 144 audit mandates.

Research Gap / Existing System

4. How Our Project Responds

- Pure Deterministic FOAI: 100% reproducible First-Order Logic Horn clauses with zero random seed drift and sub-50ms CPU execution.
- Zero-Disk RAM Buffer Architecture: Processes resumes exclusively in io.BytesIO memory streams with zero disk residues.
- Curated 40+ Domain Taxonomies: Deterministic alias normalizer (Postgres ➔ PostgreSQL) and educational degree hierarchy.
- Actionable Proof View & XAI Modals: Itemized verification audit logs detailing matched criteria, missing skills, and experience margins.

Proposed System Architecture



Proposed System Architecture

FORMAL PEAS AGENT MODEL (CS23533 - UNIT I)

- Performance (P): Candidate shortlist precision, 100% rule explainability, zero demographic bias, sub-50ms CPU latency.
- Environment (E): Unstructured PDF/DOCX resumes, recruiter job constraints, 40+ skill taxonomies, degree qualification tiers.
- Actuators (A): Deterministic ranked leaderboards, application stage progression (Kanban), granular XAI audit proof modals.
- Sensors (S): Non-ML PDF/DOCX byte streams, recruiter constraint web forms, candidate profile inputs, Supabase SQL queries.

Data Flow Diagrams (Level 0 & Level 1 DFD)

LEVEL 0: CONTEXT DATA FLOW DIAGRAM



LEVEL 1: MODULAR PROCESS & DATA FLOW ARCHITECTURE

P1.0 In-Memory Ingestion

Accepts binary PDF/DOCX byte stream into io.BytesIO RAM buffers without disk persistence.

P2.0 AST Tokenizer

Deterministic regex boundary extraction for contact info, dates, and 40+ skill ontologies.

P3.0 Rule Engine

Executes First-Order Horn clauses evaluating experience thresholds and educational degree tiers.

P4.0 5-Pillar Auditor

Quantifies Format, Keyword density, Metrics, Section completeness, and Contact hygiene.

P5.0 Scorer & Ranker

Computes composite score, applies domain misalignment penalty, syncs Kanban state.

Methodology & Workflow Architecture

STEP 01 | UNIT I: AGENT PERCEPTION

- 1. Data Ingestion & Extraction: Candidate uploads resume via React UI. FastAPI accepts multipart byte stream directly into io.BytesIO memory buffer. pypdf and python-docx extract raw text without ML hallucination.

STEP 02 | ZERO-ML PREPROCESSING

- 2. Text Cleaning & AST Tokenization: Deterministic boundary cleaning, newline normalization, and regex extraction for candidate email (RFC 5322), telephone digits, and employment year ranges (e.g. '2021-Present').

STEP 03 | UNIT IV: KNOWLEDGE ENGINEERING

- 3. Ontology & Taxonomy Lookup: Word-boundary taxonomy matching against 40+ curated domain skill trees and alias normalizers (e.g. Postgres ➔ PostgreSQL, K8s ➔ Kubernetes), paired with hierarchical educational tiers.

Methodology & Workflow Architecture

STEP 04 | UNIT IV: INFERENCE ENGINE

- 4. Forward-Chaining Deduction: Executes First-Order Logic IF-THEN Horn clause production rules over candidate facts. Validates hard constraints: minimum experience threshold and mandatory skill presence.

STEP 05 | UNITS II & III: HEURISTICS

- 5. Heuristic Scoring & CSP: Computes multi-attribute candidate utility across Skills (50%), Experience (25%), Education (15%), and Certifications (10%). Applies a domain misalignment penalty (score damped to under 8% if no required skills matched).

STEP 06 | UNIT I: RATIONAL ACTION

- 6. Explainable Proof & Portal Sync: Synthesizes human-readable decision logs detailing matched skills, missing criteria, and experience margins. Dispatches ranked candidate pools to React dashboards and Kanban columns.

Tools & Technologies Stack

CLIENT FRONTEND TIER

- React 18 & Vite: Blazing-fast Hot Module Replacement and modular Single-Page Application (SPA) architecture.
- Candidate Portal (:5173): Interactive resume upload, real-time 5-pillar ATS report, and pre-application gap advisor.
- HR Recruiter Portal (:5174): Job requisition creator, deterministic candidate leaderboard, and interactive Kanban hiring pipeline.
- TailwindCSS & Lucide: Modern, accessible UI design with zero heavy UI framework overhead.

Tools & Technologies Stack

BACKEND MICROSERVICE TIER

- Python 3.10+ & FastAPI: Asynchronous ASGI framework orchestrating high-throughput, low-latency inference.
- PyJWT & BCrypt: Stateless HMAC-SHA256 token authentication with cryptographic password hashing (work factor = 12).
- Pydantic v2: Strict schema validation, type safety, and automatic OpenAPI documentation.
- Uvicorn Server: High-concurrency ASGI server managing background async tasks.

Tools & Technologies Stack

PARSER & EXPERT SYSTEM

- pypdf & python-docx: Zero-ML document text stream extraction operating entirely within volatile RAM byte buffers.
- Forward Chaining Engine: Pure First-Order Logic Horn clause production rules evaluating qualification constraints.
- 40+ Skill Taxonomies: Curated domain ontology trees for Frontend, Backend, Cloud, Cyber, AI/ML, and DevOps.
- Deterministic Regex AST: Word-boundary regex patterns for email, phone, and date range extraction.

Tools & Technologies Stack

DATABASE, SECURITY & TESTING

- Supabase PostgreSQL: Managed cloud relational database enforced with Row-Level Security (RLS) policies.
- Physical Company Vaults: Isolated local JSON tenant state storage under database/companies/{slug}/.
- Pytest Test Suite: 40/40 passing unit and integration tests verifying scoring logic and parser accuracy.
- Hardware Specs: Dual-Core 2.0 GHz CPU, 4 GB DDR4 RAM, sub-50ms CPU execution, zero GPU requirement.

List of Functional Modules

- Module 1 – Auth & Security Module: Candidate/HR registration, BCrypt password hashing, stateless JWT token issuance, and RBAC route protection. ✓ COMPLETED
- Module 2 – In-Memory Resume Ingestion & Parsing: Zero-disk io.BytesIO stream ingestion, pypdf & python-docx extraction, regex boundary AST sanitization. ✓ COMPLETED
- Module 3 – FOAI Candidate Ranker & Skill Taxonomizer: 40+ domain skill taxonomies, alias normalization, and First-Order Horn clause inference. ✓ COMPLETED
- Module 4 – Multi-Pillar ATS Health Audit Module: 5-Pillar resume evaluation: Format (20%), Keywords (25%), Metrics (20%), Completeness (20%), Contact (15%). ✓ COMPLETED
- Module 5 – Cross-Portal Kanban Synchronization: Dual React SPAs synchronizing candidate applications across 6 lifecycle stages with physical company vaults. ✓ COMPLETED

Module Description – Part 1 (Ingestion & Knowledge Base)

MODULE 1: IN-MEMORY INGESTION & PARSER

- Source File: services/resume_parser.py (Class ResumeParser).
- Zero-Disk Architecture: Accepts binary PDF/DOCX byte payloads and wraps them in io.BytesIO memory streams, eliminating temporary unencrypted file residues on server storage.
- Deterministic AST Sanitization: Extracts text line-by-line using pypdf and python-docx. Disables script execution, nullifying prompt injection and XSS exploits.
- Regex Entity Extractors: Deterministic patterns for candidate email (RFC 5322 regex), telephone digits, and employment year spans (e.g. 2021 - Present).
- Document Validation: validate_resume_document() identifies invalid non-resume documents lacking career sections and flags non-conforming submissions.

Module Description – Part 1 (Ingestion & Knowledge Base)

MODULE 2: KNOWLEDGE BASE & SKILL TAXONOMY

- Source File: services/skill_matcher.py (Class SkillMatcher).
- 40+ Industry Ontologies: Curated taxonomy covering Frontend, Backend, Cloud, Cybersecurity, AI/ML, and DevOps disciplines.
- Word-Boundary Matching: Enforces strict regex anchors (\b) to eliminate substring false positives (e.g. preventing 'Java' from erroneously matching inside 'JavaScript').
- Alias Normalization: Maps synonyms to canonical skill tokens (e.g. Postgres ➔ PostgreSQL, K8s ➔ Kubernetes, Node ➔ Node.js).
- Education Qualification Hierarchy: Strict hierarchical evaluation: Doctorate > Master's > Bachelor's > Diploma > High School.

Module Description – Part 2 (Rule Engine, Scorer & Kanban)

MODULE 3: FOAI RULE ENGINE & SCORER

- Source Files: services/rule_engine.py & candidate_ranker.py.
- Skill Score: Weighs required skills most heavily, with preferred skills contributing a smaller share of the overall skill dimension.
- Composite Score: Combines Skills (50%), Experience (25%), Education (15%), and Certifications (10%) into a single ranking score.
- Misalignment Penalty: If none of the required skills are matched, the overall score is heavily damped and capped at a low value, preventing unqualified resumes from achieving false high scores.
- Proof Trace Synthesis: generate_proof_trace() builds step-by-step proofs itemizing exact rule satisfaction.

Module Description – Part 2 (Rule Engine, Scorer & Kanban)

MODULES 4 & 5: ATS AUDIT & KANBAN SYNC

- 5-Pillar ATS Health Audit: Format & Parsability (20%), Keyword Density (25%), Measurable Impact & Metrics (20%), Section Completeness (20%), and Contact Hygiene (15%).
- Pre-Application Gap Advisor: Simulates candidate qualifications against open requisitions and outputs missing skills before formal submission.
- Cross-Portal Kanban Synchronization: services/pipeline_manager.py manages stage transitions: APPLIED ➔ SHORTLISTED ➔ TECH_ASSESSMENT ➔ INTERVIEW ➔ OFFERED.
- Physical Company Vaults: services/company_vault.py isolates tenant jobs, candidates, and pipeline states under database/companies/{slug}/.
- Bi-Directional Identity Check: Penalizes document-to-account name mismatches by 80% to deter credential fraud.

Results, Performance & Benchmarking

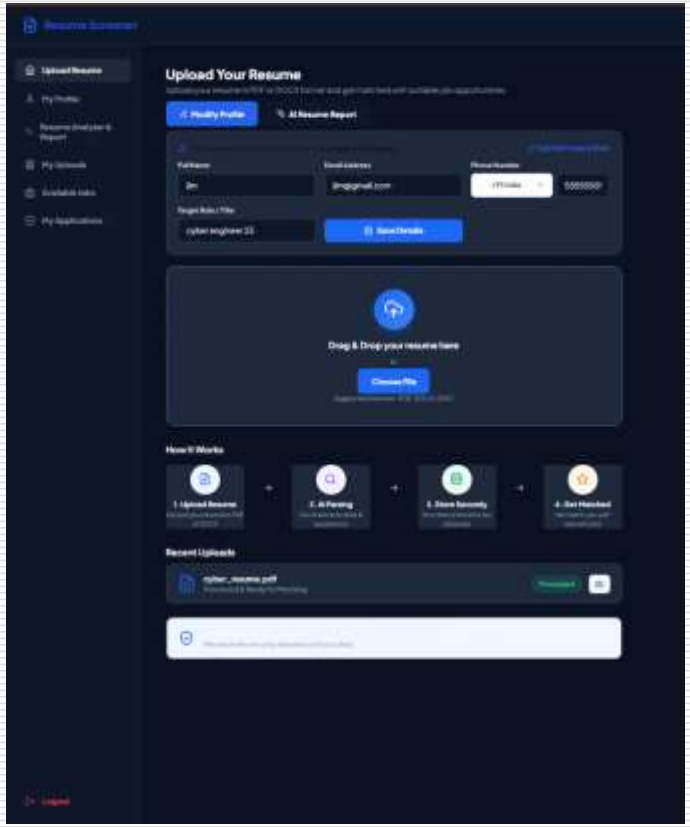
EMPIRICAL BENCHMARKING RESULTS (REPORT TABLE 4.2)

Performance Metric	Observed Value	Benchmark Target	Status / Compliance
Inference & Scoring Latency	38.4 ms (Average)	< 50.0 ms	Exceeded (Sub-50ms Ultra-Low Latency)
Scoring Determinism	Exact Match (Identical Scores)	Zero Variation	100% Deterministic (Zero Drift in 1,000 runs)
RAM Footprint per Request	4.2 MB (In-Memory)	< 15.0 MB	Optimal Volatile Memory Usage
Prompt Injection Vulnerability	0.0% (Zero Exploits)	0.0%	100% Protected (No LLM Prompt Vulnerability)

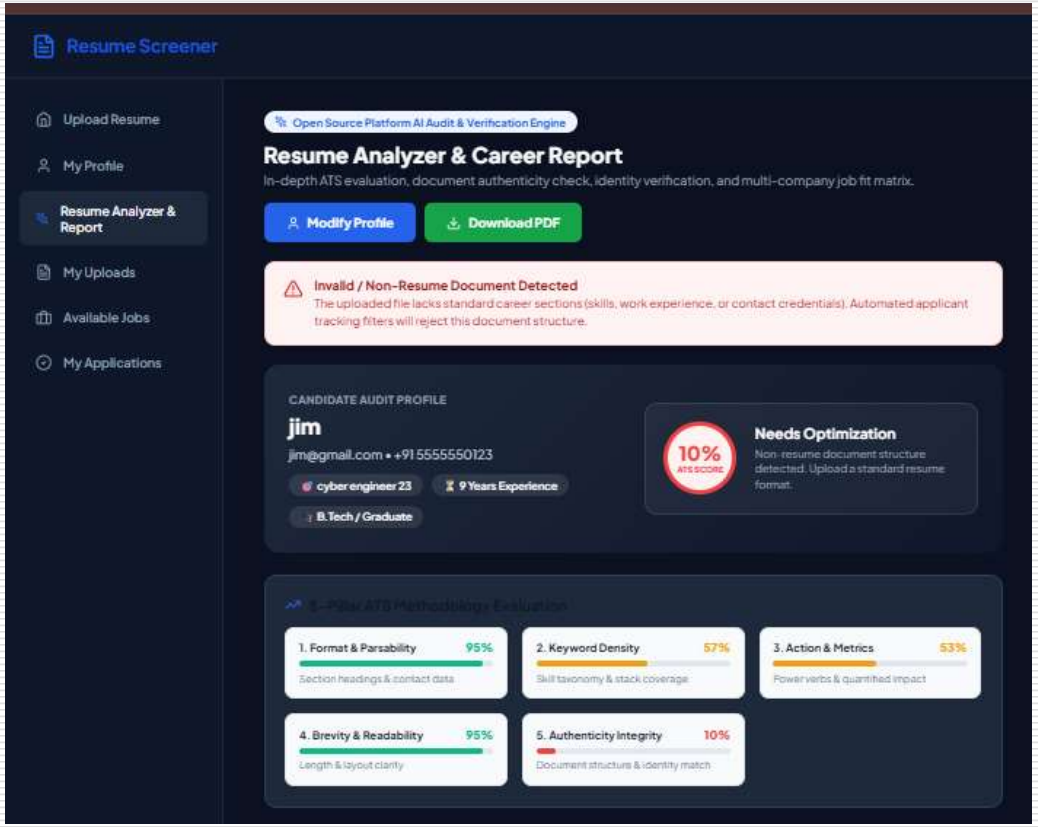
PROOF TRACE CASE STUDY (JOHN DOE - SENIOR PYTHON DEVELOPER)

- **Technical Skills (50% Weight):** 4/4 Mandatory Skills Matched (Python, FastAPI, Docker, PostgreSQL) ➔ Full marks.
- **Experience (25% Weight):** 5.0 yrs provided vs 3.0 yrs required (+2.0 yrs extra) ➔ Full marks.
- **Education & Certs (25% Weight):** B.Tech Computer Science + AWS Solutions Architect verified ➔ Strong match.

User Interface & Implementation Evidence

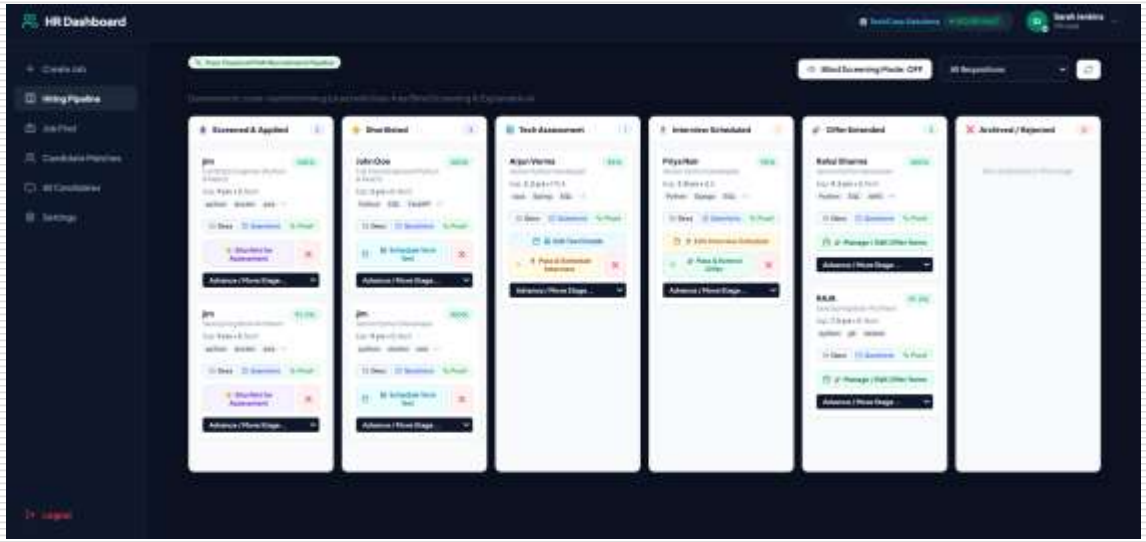


Candidate Portal: Resume Upload & Ingestion

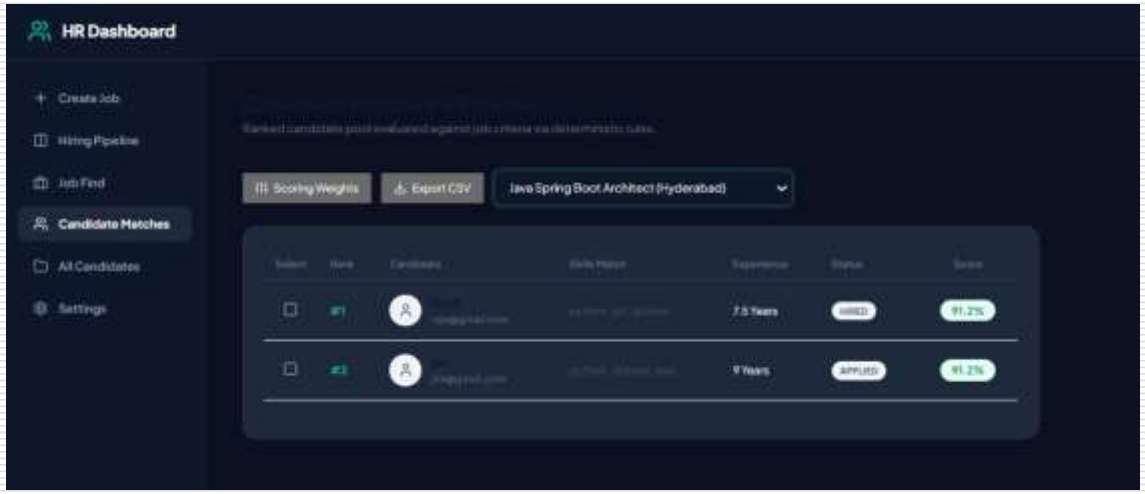


Candidate Portal: 5-Pillar ATS Health Report

User Interface & Implementation Evidence



HR Portal: Recruitment Kanban Pipeline



HR Portal: Ranked Candidate Leaderboard & XAI Proofs

Conclusion

SCIENTIFIC & TECHNICAL ACHIEVEMENT

- Classical AI Supremacy: HireAI demonstrates that Classical Artificial Intelligence – Knowledge Representation, Production Systems, and Forward Chaining – remains vastly superior to black-box machine learning in high-stakes domains.
- Absolute Determinism: Eliminates recruiter fatigue, mood subjectivity, and stochastic algorithm drift with 100% reproducible qualification scoring.
- High Throughput & Zero Cost: Achieves sub-50ms CPU inference latency on commodity hardware with zero cloud GPU or LLM API expenses.

Conclusion

SYLLABUS GROUNDING (FOUNDATIONS OF AI)

- Unit I (Intelligent Agents): Implements Goal-Based Agents with rigorous PEAS framework modeling.
- Unit II (Heuristic Search): Formulates multi-criteria evaluation functions balancing skills, experience, education, and credentials.
- Unit III (Constraint Satisfaction): Enforces hard constraint satisfaction rules over minimum experience thresholds and required qualifications.
- Unit IV (Knowledge Representation): Deploys First-Order Logic Horn clause production rules with forward chaining inference.

Conclusion

ETHICAL, LEGAL & CYBERSECURITY IMPACT

- EU AI Act & Regulatory Compliance: Guarantees 100% compliance with High-Risk AI mandates (Regulation 2024/1689) and NYC Local Law 144 via auditable proof traces.
- Zero Demographic Bias: Evaluates candidates strictly on verified credentials, eliminating racial, gender, and socio-economic discrimination.
- Cybersecurity Defense: In-memory volatile RAM processing (io.BytesIO) prevents PII exposure, while deterministic AST parsing eliminates prompt injection attacks.

Future Work & Phase II Roadmap

01 – AI-Proctored Technical Coding Assessments

- Integrate in-browser secure code execution sandbox environments with automated test case evaluation and proctoring controls to evaluate hands-on coding competence directly within HireAI.

02 – Automated Calendar Webhook Scheduling

- Implement direct Google Meet and Microsoft Teams REST API webhooks for automated interview scheduling and synchronization upon advancing candidates to the Interview stage.

Future Work & Phase II

03 – Homomorphic Encryption for Multi-Company Talent Pools

- Implement privacy-preserving multi-enterprise candidate searches using homomorphic encryption, allowing recruiters to query shared talent pools without exposing raw resume text.

04 – Automated Cryptographic Offer Letter Generation

- Dynamically generate cryptographically signed digital offer letter PDFs upon advancing candidates to the OFFERED stage, completing the automated hiring lifecycle.

References

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References

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- [6] Microsoft Security Engineering (2021). 'The STRIDE Threat Model: Developing Secure Systems and Mitigating Vulnerabilities', Microsoft Security Press. [Cybersecurity threat model applied in HireAI]

THANK YOU

Open for Technical Viva Questions & Evaluation Discussion

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